



For the Partial Fulfillment of the
Requirements of Software Engineering 2

SOFTWARE REQUIREMENTS SPECIFICATIONS

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1. Introduction

a. (i) Purpose of this Document –

The purpose of this document is to formally specify the requirements for the Government Appointment Booking System. It serves as the guide for the development team, stakeholders, and quality assurance team to ensure that the system meets the intended objectives. The document provides clarity on the system's scope, objectives, and functionalities, ensuring alignment between developers and users.

b. (ii) Scope of this document –

The document outlines the overall working of the Government Appointment Booking System, its objectives, and the value it provides to the stakeholders and users. The system automates the scheduling process, improves the accessibility, and streamlines the application process for government applications.

Specific: The system shall deliver a fully functional, web-based appointment booking platform supporting four distinct user roles; Client, Staff, Manager, and Administrator each with role-specific dashboards, access controls, and capabilities covering booking, processing, tracking, and account management

Measurable: The system shall reduce appointment scheduling errors and double bookings to zero through automated slot allocation and real-time calendar synchronization, and shall successfully deliver automated email notifications for 100% of pending, confirmed, completed, or cancelled appointments

Achievable: The system shall be developed and deployed within a 3-month timeline by the designated project team, utilizing only approved and cost-constrained tools and technologies, while fulfilling all seven functional modules defined in this requirements specification.

Relevant: The system shall replace manual and walk-in scheduling processes for Filipino citizens accessing government services, improving public accessibility by providing a 24/7 online self-service platform that reduces physical congestion, communication delays, and administrative workload for government agency staff.

Time-bound: A fully tested and operational version of the Government Appointment Booking System shall be completed and ready for deployment within 3 months from the official project start date, with all functional requirements validated by the quality assurance team prior to release.



c. (iii) **Overview –**

The Appointment Booking System is a web-based application designed to streamline appointment scheduling and management for government applications. It enables clients (Filipino citizens) to book appointments, track status, and receive notifications, while staff and administrators manage requests, schedules, and services. The system replaces manual scheduling methods, reducing conflicts, delays, and inefficiencies. It provides real-time updates, secure authentication, and centralized record-keeping, improving communication and operational efficiency.

2. General description

The Appointment Booking System is a web-based application that automates appointment scheduling, confirmation, and tracking while supporting three user roles: Admin, Staff, and Client. It offers secure user authentication with role-based access control, calendar-integrated appointment booking with real-time availability, and comprehensive management of appointment requests, including confirmations, rescheduling, and cancellations. The system also facilitates client and staff account management, service definition for appointment types, and sends notifications for confirmations, updates, and reminders. Additionally, it maintains audit logs and generates reports for administrative oversight, enhancing operational efficiency and user experience.

The users of the Appointment Booking System include *clients* (Filipino citizens) who can easily book appointments, check availability, and track the status of their requests; *government agency staff* that can efficiently manage appointment requests, update statuses, and communicate with clients; and *administrators* who oversee system activities, manage users and services, and generate reports to ensure smooth operation and oversight.

The Appointment Booking System offers several key features and benefits including an online self-service appointment booking, automated slot allocation and scheduling, real-time calendar view of available appointment slots, automated email notifications for confirmations and updates, and role-based dashboards tailored for *administrators*, *government agency staff*, and *clients*. It ensures secure authentication with password complexity requirements and session management, provides appointment history and real-time status tracking, and maintains audit logs to support accountability and transparency within the system. communicate with clients.

The Appointment Booking System addresses inefficiencies inherent in manual scheduling processes by reducing communication delays and ensuring data integrity. It enhances the overall user experience by providing 24/7 access to appointment booking and status checking, while also improving staff resource allocation and administrative oversight, thereby contributing to more efficient operations. Overall, the system will provide a benefit of reducing



manual processing and walk-in congestion, offering a faster, accessible, and more convenient way to access public services.

3. Functional Requirements

The functional requirements of the Government Appointment Booking System are as follows:

Module 1: Login Window

- **FR 1.1** The system shall allow users (admin-level, manager-level, staff-level, and client) to log in using their registered email address and password.
- **FR 1.2** The system shall enforce Role-Based Access Control (RBAC) to restrict access to specific features and dashboards based on the user's assigned role.
- **FR 1.3** The system shall allow users to securely terminate their session through logout.
- **FR 1.4** The system shall enforce password complexity requirements (min 8 chars, mix of cases, numbers), and prevent duplicate accounts.
- **FR 1.5** The system shall prevent unauthorized access when invalid credentials are provided.
- **FR 1.6** The system shall provide a secure password recovery mechanism via an email reset link for users who forget their credentials.
- **FR 1.7** The system shall allow new users to register a client-level account.

Module 2: Booking Window

- **FR 2.1** The system shall allow client users to view a calendar interface that displays real-time availability of appointment slots based on the selected government service, and let them create an appointment.
- **FR 2.2** The system shall prevent multiple client users from booking the exact same capacity-limited time slot to avoid double booking
- **FR 2.3** The system shall require users to complete a digital booking form, submitting all required personal details and necessary documentation for their selected service.
- **FR 2.4** The system shall generate a unique, alphanumeric Appointment Tracking Number for every successfully submitted appointment request.
- **FR 2.5** The system shall send an email notification upon successful booking containing the Appointment Tracking Number, date, time, and service details to the user's email address.
- **FR 2.6** The system shall store all appointment records in the database for retrieval and processing.
- **FR 2.7** The system shall allow client users to upload required documents during the booking process, supporting common file



formats (e.g., PDF, JPG, PNG), with a maximum file size limit per upload.

Module 3: Appointment Processing Window

- **FR 3.1** The system shall provide staff and manager-level accounts with a dashboard to retrieve and view the appointment information from the appointments.
- **FR 3.2** The system shall allow staff-level and manager-level users to update appointment details and status (e.g., pending, confirmed, completed).
- **FR 3.3** The system shall prevent staff-level and manager-level users from processing appointments outside configured service operating hours.
- **FR 3.4** The system shall allow staff to add remarks to an appointment record for a better context of the client's status.
- **FR 3.5** The system shall generate an update notification containing the new status of a user's appointment and send an email notification.
- **FR 3.6** The system shall archive completed appointments in the database
- **FR 3.7** The system shall log all appointment processing actions into a centralized Audit Log.
- **FR 3.8** The system shall allow staff-level and manager-level users to view and download documents uploaded by the client for a given appointment record.

Module 4: Appointment Tracking Window

- **FR 4.1** The system shall provide clients with an interface to monitor their existing appointments.
- **FR 4.2** The system shall retrieve and display the real-time status of a tracked appointment upon request.
- **FR 4.3** The system shall allow clients to view their appointment history
- **FR 4.4:** The system shall display a read-only historical timeline of appointment status changes for transparency.
- **FR 4.5** The system shall allow clients to filter and sort their appointment list by date, status, or service type.
- **FR 4.6** The system shall display upcoming appointments separately from past appointments for easier navigation.

Module 5: Client Account Management Window

- **FR 5.1** The system shall allow clients to view their personal profile information.
- **FR 5.2** The system shall allow clients to update their personal profile information.
- **FR 5.3** The system shall allow clients to deactivate their accounts.
- **FR 5.4** The system shall allow clients to update their password
- **FR 5.5** The system shall validate and save changes to the users account



- **FR 5.6** The system shall log the changes made by the user to their account

Module 6: Manager and Staff Account Management

- **FR 6.1** The system shall allow admin-level users the exclusive ability to create, view, update, and archive manager-level and staff-level accounts
- **FR 6.2** The system shall allow manager-level users to assign staff-level accounts to a specific service for operational responsibilities
- **FR 6.3** The system shall validate and save changes made to the associated account
- **FR 6.4** The system shall log all actions involving the creation or modification of manager and staff accounts into a centralized Audit Log
- **FR 6.5** The system shall validate that no accounts are duplicated

Module 7: Service Management Window

- **FR 7.1:** The system shall allow a manager-level user to create a new service, defining the service name, description, available operating days, business hours, maximum concurrent slot capacity, government service department, and required client documents, if applicable
- **FR 7.2** The system shall allow a manager-level user to view all existing services and their parameters
- **FR 7.3** The system shall allow a manager-level user to modify existing service and their parameters
- **FR 7.4** The system shall allow a manager-level user to temporarily suspend or archive existing service, automatically removing its availability from the public calendar
- **FR 7.5** The system shall log all actions involving the creation, modification, and archiving of services into a centralized Audit Log
- **FR 7.6** The system shall validate that no services are duplicated

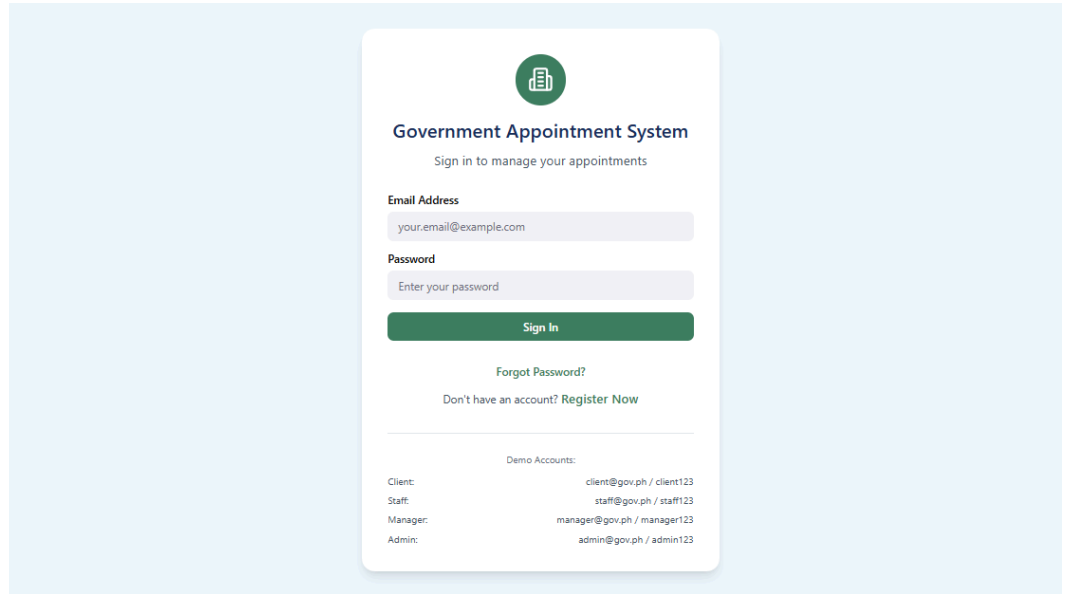
4. Interface Requirements

This section describes how the Government Appointment Booking System interacts with users, external systems, and internal components through defined interfaces. These interfaces enable communication using structured data formats, protocols, messages, and user interactions, ensuring efficient and secure data exchange.

4.1 User Interface (UI)

The system shall provide a web-based graphical user interface accessible through modern browsers.

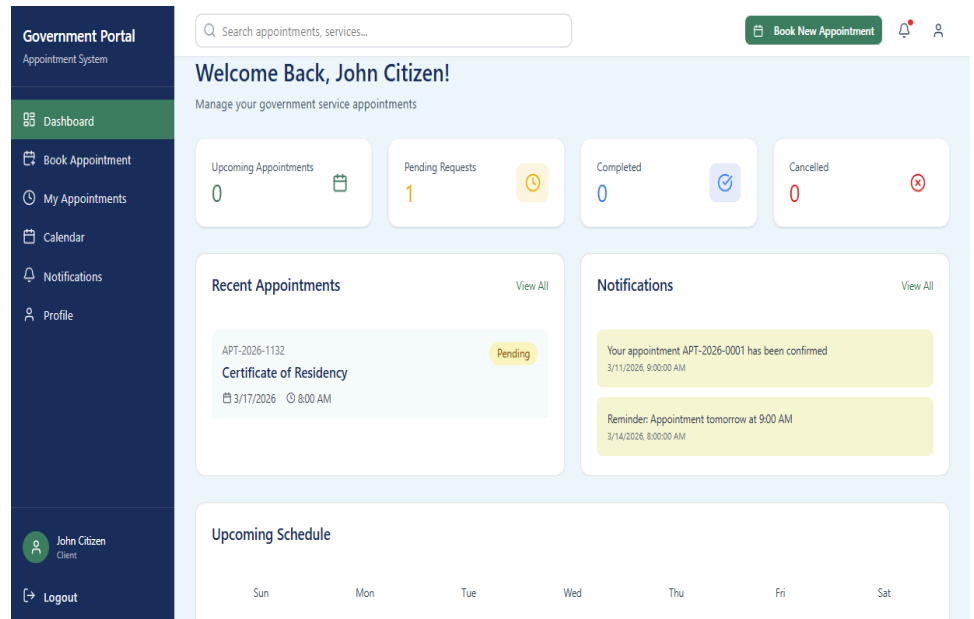
i. Login/Auth Page



The login page features a central white card on a light blue background. At the top of the card is a green circular icon with a white document symbol. Below this is the title "Government Appointment System" and the subtitle "Sign in to manage your appointments". The form includes two input fields: "Email Address" with the placeholder "your.email@example.com" and "Password" with the placeholder "Enter your password". A green "Sign In" button is positioned below the password field. Below the button are links for "Forgot Password?" and "Don't have an account? Register Now". At the bottom, a "Demo Accounts:" section lists four roles with their respective email addresses: Client (client@gov.ph / client123), Staff (staff@gov.ph / staff123), Manager (manager@gov.ph / manager123), and Admin (admin@gov.ph / admin123).

ii. Client Interface

1. Client Dashboard



The dashboard is titled "Government Portal Appointment System". It features a dark blue sidebar with navigation links: Dashboard, Book Appointment, My Appointments, Calendar, Notifications, and Profile. The main content area has a search bar and a "Book New Appointment" button. A welcome message "Welcome Back, John Citizen!" is displayed. Below this, there are four summary cards: "Upcoming Appointments" (0), "Pending Requests" (1), "Completed" (0), and "Cancelled" (0). The "Recent Appointments" section shows a pending appointment for "Certificate of Residency" on 3/17/2026 at 8:00 AM. The "Notifications" section shows a confirmation for appointment APT-2026-0001 and a reminder for tomorrow at 9:00 AM. The "Upcoming Schedule" section shows a weekly calendar view from Sunday to Saturday.

2. Appointment Booking



Government Portal
Appointment System

Dashboard

Book Appointment

My Appointments

Calendar

Notifications

Profile

John Citizen
Client

Logout

Search appointments, services...

Book New Appointment

Book an Appointment

Schedule your appointment with government services

1
Category

2
Service

3
Date & Time

4
Details

Select Service Category

Certifications & Documents
General government document requests and certifications

Public Safety
Documents issued by agencies such as the Philippine National Police and Bureau of Fire Protection

Health & Social Welfare
Programs and records handled by the Department of Health and Department of Social Welfare and Development

Civil Registry
Official personal records issued by the Philippine Statistics Authority

Disaster & Community Services
Services handled by Local Government Units (LGUs) and National Disaster Risk Reduction and Management Council

< Back

Next >

Government Portal
Appointment System

Dashboard

Book Appointment

My Appointments

Calendar

Notifications

Profile

John Citizen
Client

Logout

Search appointments, services...

Book New Appointment

Book an Appointment

Schedule your appointment with government services

1
Category

2
Service

3
Date & Time

4
Details

Select a Service

General government document requests and certifications

Certificate of Residency
Government service request

Business Permit Request
Government service request

Community Tax Certificate (Cedula)
Government service request

Barangay Clearance for Business
Government service request

< Back

Next >

Page 8 of 33



Government Portal
Appointment System

Dashboard

Book Appointment

My Appointments

Calendar

Notifications

Profile

John Citizen
Client

Logout

Search appointments, services...

Book New Appointment

Book an Appointment

Schedule your appointment with government services

1

Category

2

Service

3

Date & Time

4

Details

Choose Date & Time

Select Date

March 2026

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

* Weekends are not available for appointments

Select Time Slot

8:00 AM

9:00 AM

10:00 AM

11:00 AM

1:00 PM

2:00 PM

3:00 PM

4:00 PM

Government Portal
Appointment System

Dashboard

Book Appointment

My Appointments

Calendar

Notifications

Profile

John Citizen
Client

Logout

Search appointments, services...

Book New Appointment

Book an Appointment

Schedule your appointment with government services

1

Category

2

Service

3

Date & Time

4

Details

Your Details

Full Name *

Juan Dela Cruz

Email Address *

juan@example.com

Phone Number *

09123456789

Address

City, Province

Required Documents for Certifications & Documents

Please prepare the following documents:

- Valid Government-issued ID
- Proof of Address
- Birth Certificate (for civil registry documents)
- Previous version of document (if applicable)

Click to upload or drag and drop

PDF, JPG, PNG (max 5MB)

Additional Notes

Any additional information...

Appointment Summary

Category: Certifications & Documents



Appointment Booked Successfully!

Your appointment request has been submitted and is pending approval

Tracking Number

APT-2026-6491

Save this number to track your appointment

Service

Date
N/A

Time

Office Location
City Hall, Main Office

Important Instructions

- Please wait for confirmation email before visiting the office
- Bring all required documents on your appointment date
- Arrive at least 15 minutes before your scheduled time
- Present your tracking number and valid ID upon arrival
- If you need to reschedule, please do so at least 24 hours in advance

[View My Appointments](#)

3. Appointment Tracking

My Appointments

View and manage your appointment requests

All Status

Tracking Number	Service	Date	Time	Status	Actions
APT-2026-1132	Certificate of Residency	3/17/2026	8:00 AM	Pending	View Reschedule Cancel
APT-2026-6491	Certificate of Residency	3/18/2026	8:00 AM	Pending	View Reschedule Cancel



Appointment Details

View complete information about your appointment

Tracking Number
APT-2026-1132

Service Certificate of Residency	Status Pending
Date 3/17/2026	Time 8:00 AM
Created At 3/16/2026, 1:28:33 AM	

[Download Receipt](#)[Close](#)

4. Appointment Calendar

Appointment Calendar

View all scheduled appointments

March 2026

[<](#)[>](#)

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15 9:00 AM - Birth Certifi...	16	17	18	19	20 2:00 PM - Cedula Ap...	21
22	23	24	25	26	27	28

5. Notifications



Notifications

Stay updated with your appointment status

✓ Your appointment APT-2026-0001 has been confirmed
3/11/2026, 9:00:00 AM

🕒 Reminder: Appointment tomorrow at 9:00 AM
3/14/2026, 8:00:00 AM

6. Profile

My Profile

Manage your account information



John Citizen

Client

Full Name

John Citizen

Email Address

client@gov.ph

Phone Number

09123456789

Address

City, Province

Edit Profile

Change Password



iii. Staff Interface

1. Staff Dashboard

Government Portal
Appointment System

Dashboard

Calendar

Process Requests

Notifications

Profile

Maria Staff
Staff

Logout

Search appointments, services...

Staff Dashboard - Welcome, Maria Staff!
Process and manage appointment requests

Today's Appointments
0

Pending Approval
1

Confirmed
1

Completed Today
0

Pending Appointment Requests

Tracking Number	Client Name	Service	Date	Time	Status	Actions
APT-2026-0002	John Citizen	Cedula Application	3/20/2026	2:00 PM	Pending	<button>Approve</button> <button>Reject</button>

View All

Today's Schedule

9AM Birth Certificate Request
John Citizen

Confirmed

2. Calendar

Appointment Calendar

View all scheduled appointments

March 2026

<>

Sun

Mon

Tue

Wed

Thu

Fri

Sat

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15
9:00 AM - Birth Certif...

16

17

18

19

20
2:00 PM - Cedula Ap...

21

3. Appointment Processing



Process Appointment Requests

Review and approve appointment requests

Tracking Number	Client Name	Service	Date	Time	Status	Actions
APT-2026-0001	John Citizen	Birth Certificate Request	3/15/2026	9:00 AM	Confirmed	No actions available
APT-2026-0002	John Citizen	Cedula Application	3/20/2026	2:00 PM	Pending	<button>✓ Approve</button> <button>✗ Reject</button> <button>Remarks</button>

iv. Manager Interface

1. Manager Dashboard

Government Portal
Appointment System

Dashboard

Calendar

Process Requests

Service Management

Notifications

Profile

Carlos Manager
Manager

Logout

Search appointments, services...

Manager Dashboard - Welcome, Carlos Manager!

Manage services, staff, and oversee appointment operations

Today's Appointments
0

Pending Approval
1

Confirmed
1

Completed Today
0

Pending Appointment Requests

View All

Tracking Number	Client Name	Service	Date	Time	Status	Actions
APT-2026-0002	John Citizen	Cedula Application	3/20/2026	2:00 PM	Pending	<button>Approve</button> <button>Reject</button>

Today's Schedule

gAM

Birth Certificate Request
John Citizen

Confirmed

2. Service Management



Service Management

Manage government services and availability

+ Add New Service

Birth Certificate Request

Edit Active Delete

Request for authenticated copy of birth certificate

Department
Civil Registry

Business Hours
8:00 AM - 5:00 PM

Operating Days
Monday, Tuesday, Wednesday, Thursday, Friday

Maximum Daily Slots
20

Required Documents

Valid ID Authorization Letter (if representative)

Business Permit Application

Edit Active Delete

New business permit application and renewal

Department
Business Affairs

Business Hours
8:00 AM - 5:00 PM

Operating Days
Monday, Tuesday, Wednesday, Thursday, Friday

Maximum Daily Slots
15

Required Documents

DTI Registration Barangay Clearance Valid ID

Add New Service

Create a new government service offering

Select Category *

Choose a category

Service Request Name *

e.g., Birth Certificate Request

Duration *

e.g., 30 minutes

Price *

e.g., 150

Description / Notes *

Describe the service details or requirements...

Cancel

Save Service



v. Admin Interface

1. Staff Management

Staff Management

Manage staff accounts and permissions

[+ Add New Staff](#)

Staff Name	Role	Assigned Service	Email	Status	Actions
Maria Staff	Staff	Civil Registry	staff@gov.ph	active	Edit Deactivate Delete
Carlos Manager	Manager	All Services	manager@gov.ph	active	Edit Deactivate Delete

Add New Staff

Enter the details of the new staff member.

Name

John Doe

Address

123 Main St

Role

Selec ▾

Email

email@example.com

Username

username

Temporary Password

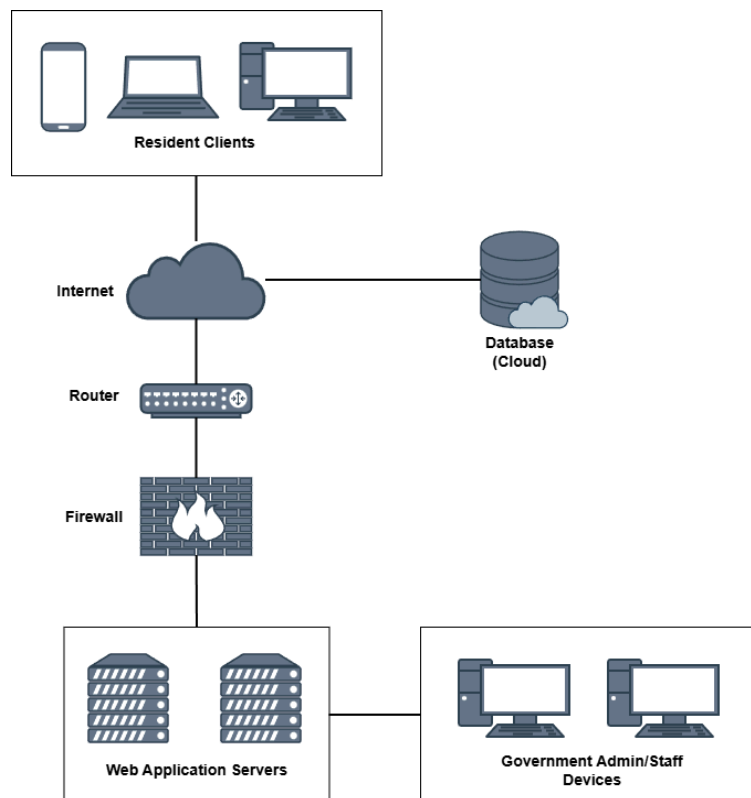
password

Add Staff



4.2 Deployment/Network Diagram

- The system facilitates communication between users and the server environment over the Internet. All data transmission across the network is strictly governed by the HTTPS protocol to ensure end-to-end encryption. The system supports real-time data updates, such as calendar availability and appointment statuses, through the use of continuous data streams.



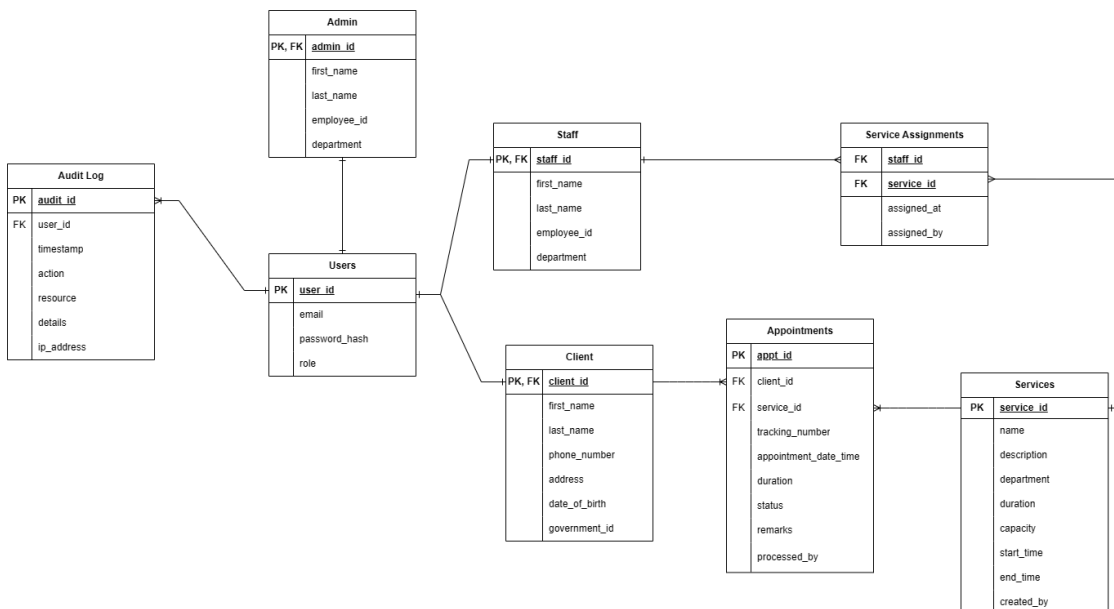


4.5 Data Interface

The system shall define how data is stored, retrieved, and managed within the system.

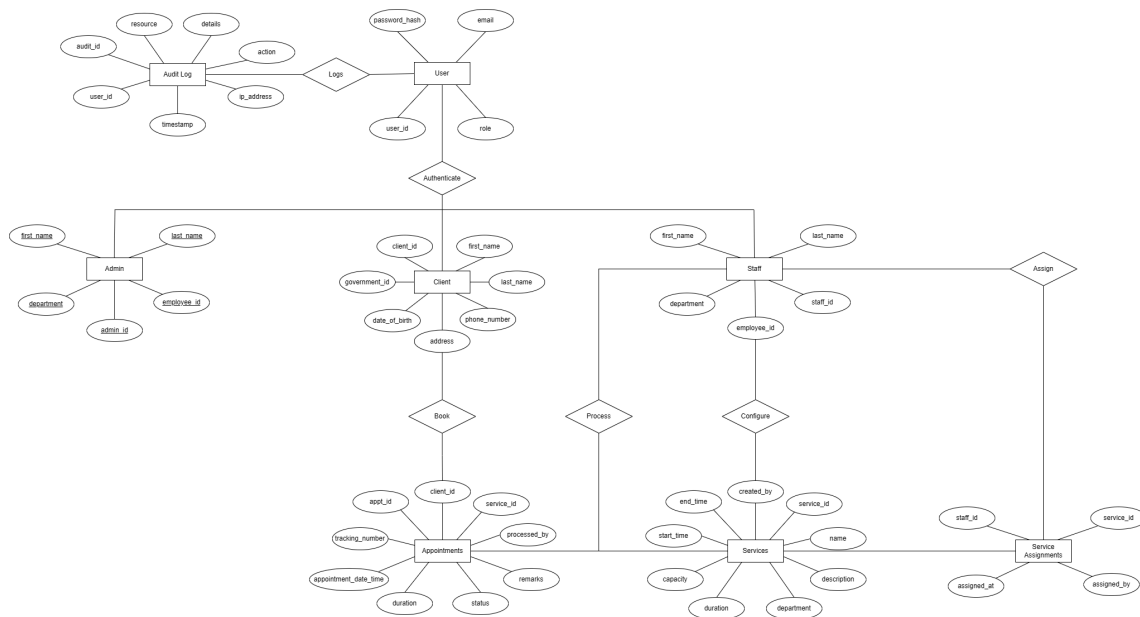
- The system shall interact with a centralized database for storing:
 - User accounts
 - Appointment records
 - Services and schedules
 - Audit Logs
- The system shall enforce data integrity and validation rules during data exchange.
- The system shall use structured queries and database relationships (e.g., foreign keys, RLS) to maintain consistency.

i. Entity Relationship Notation





ii. Entity Relationship Diagram

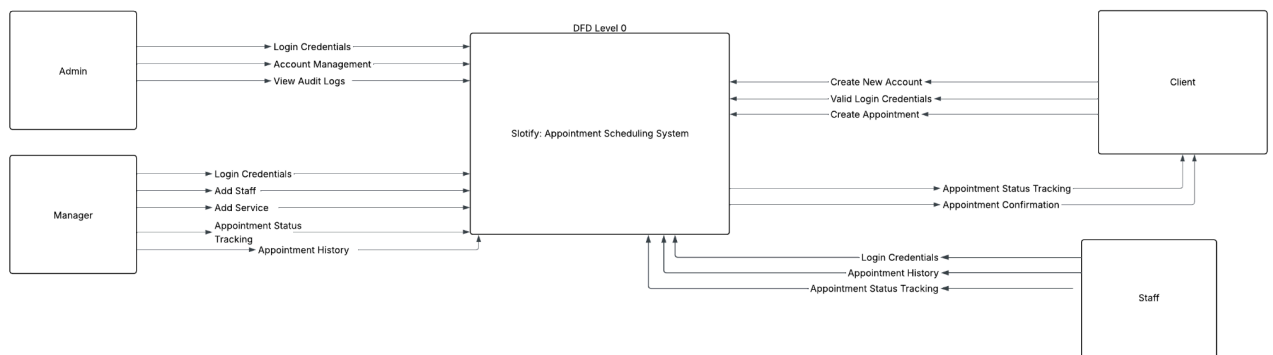




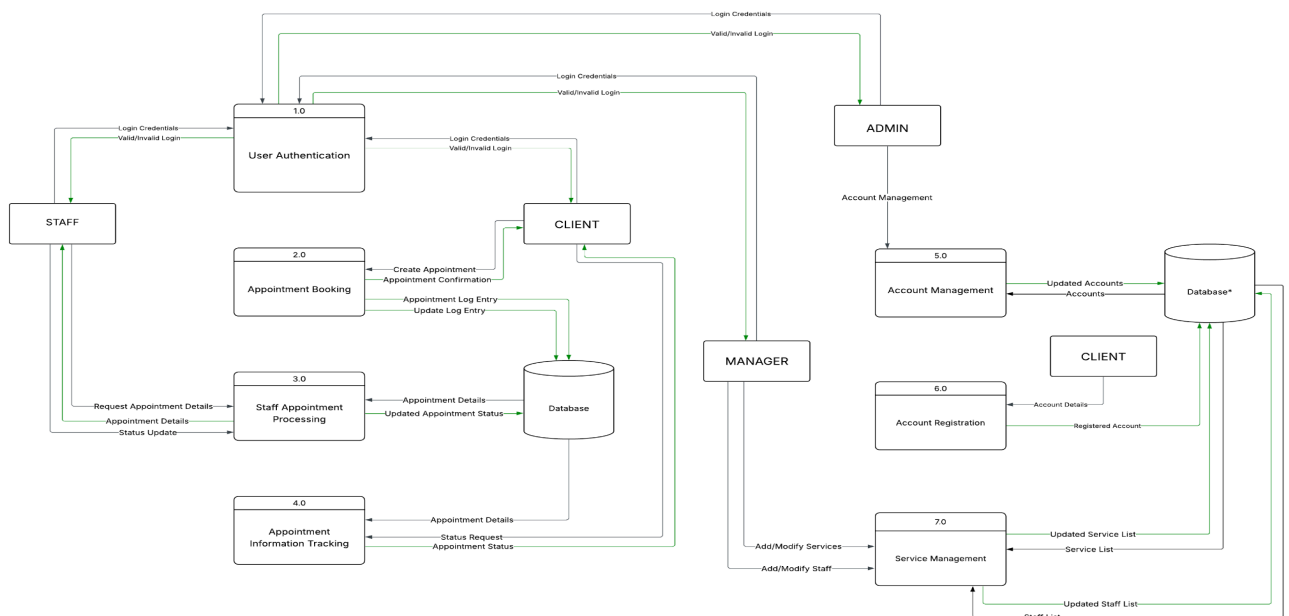
4.6 Process Diagram

- The diagram presents how data moves within the system, showing how different users interact with system components and how information is stored and retrieved from the database.
- It outlines the processes involved in handling user accounts, role-based access (client, staff, and manager), records, and system transactions, highlighting how data flows across various system functions.

i. Data Flow Diagram Level 0

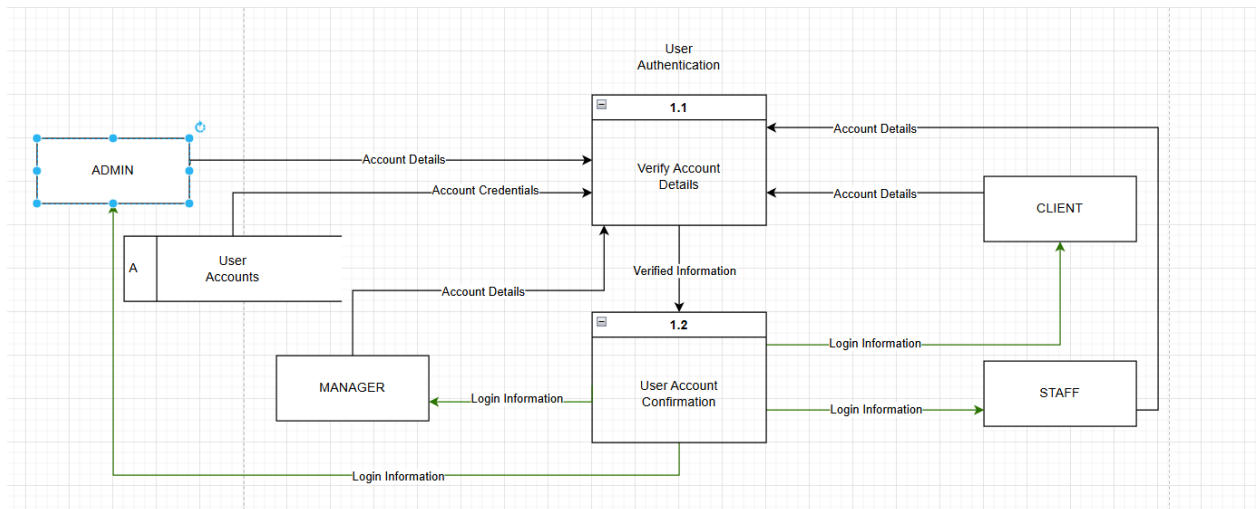


ii. Data Flow Diagram Level 2

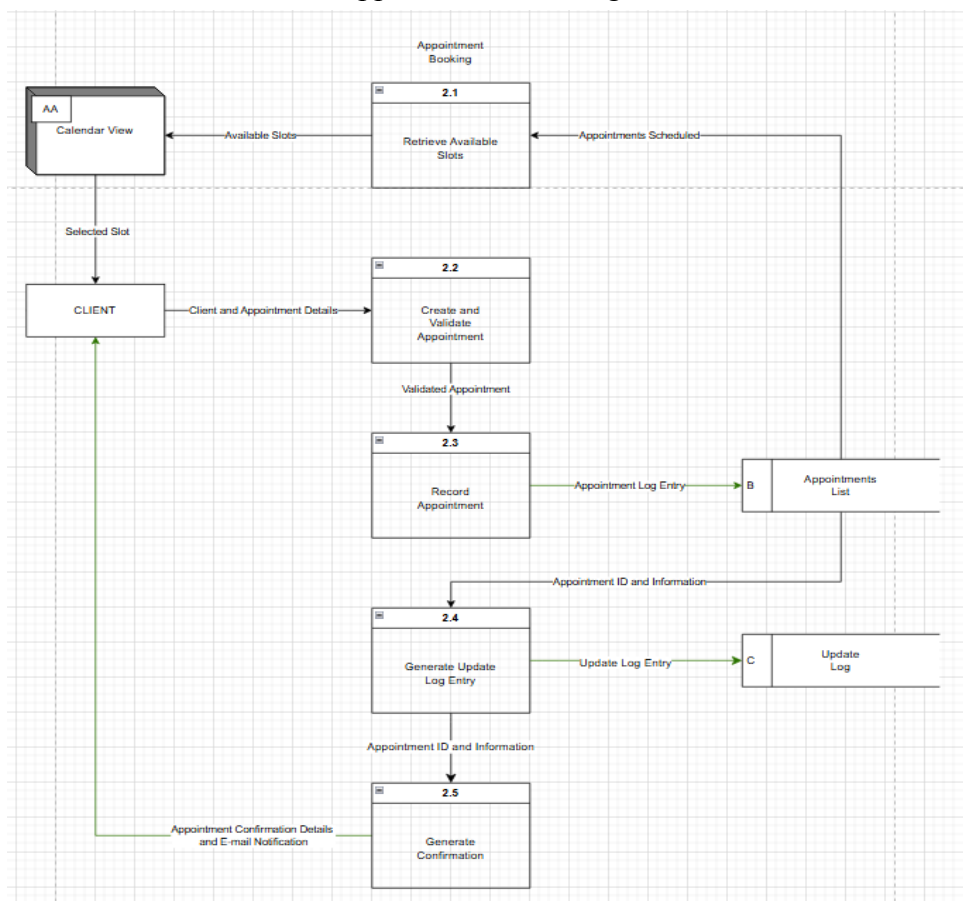


iii. Data Flow Diagram Level 2

- Module 1: User Authentication

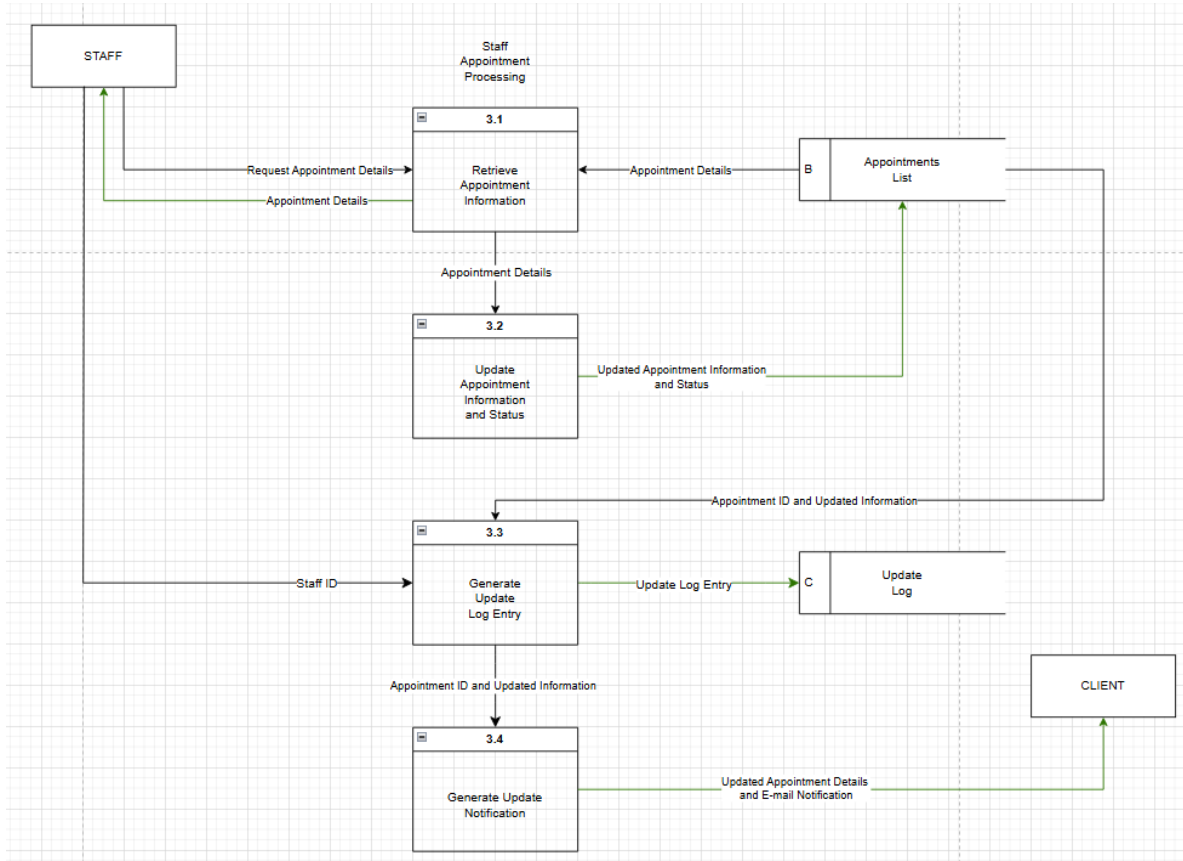


- Module 2: Appointment Booking

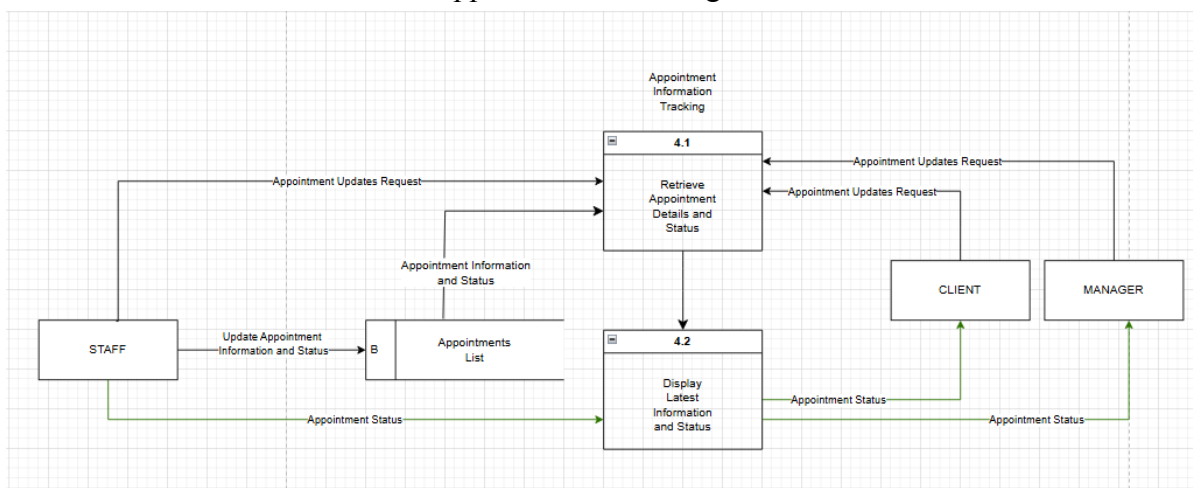




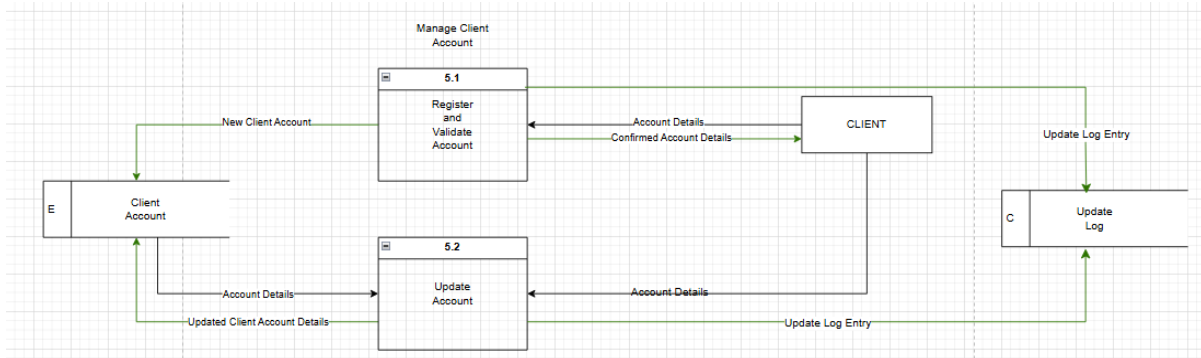
- Module 3: Staff Appointment Processing



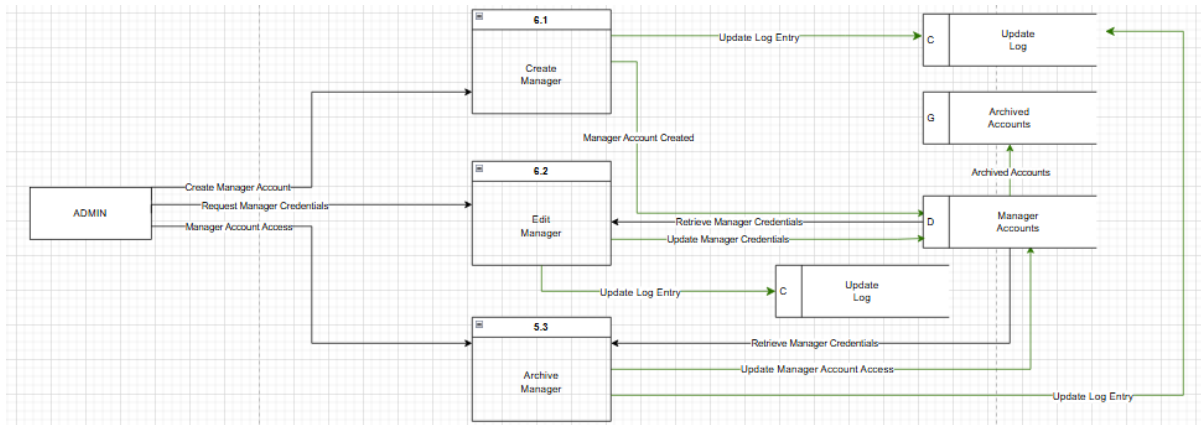
- Module 4: Appointment Tracking Window



- Module 5: Client Account Management Window

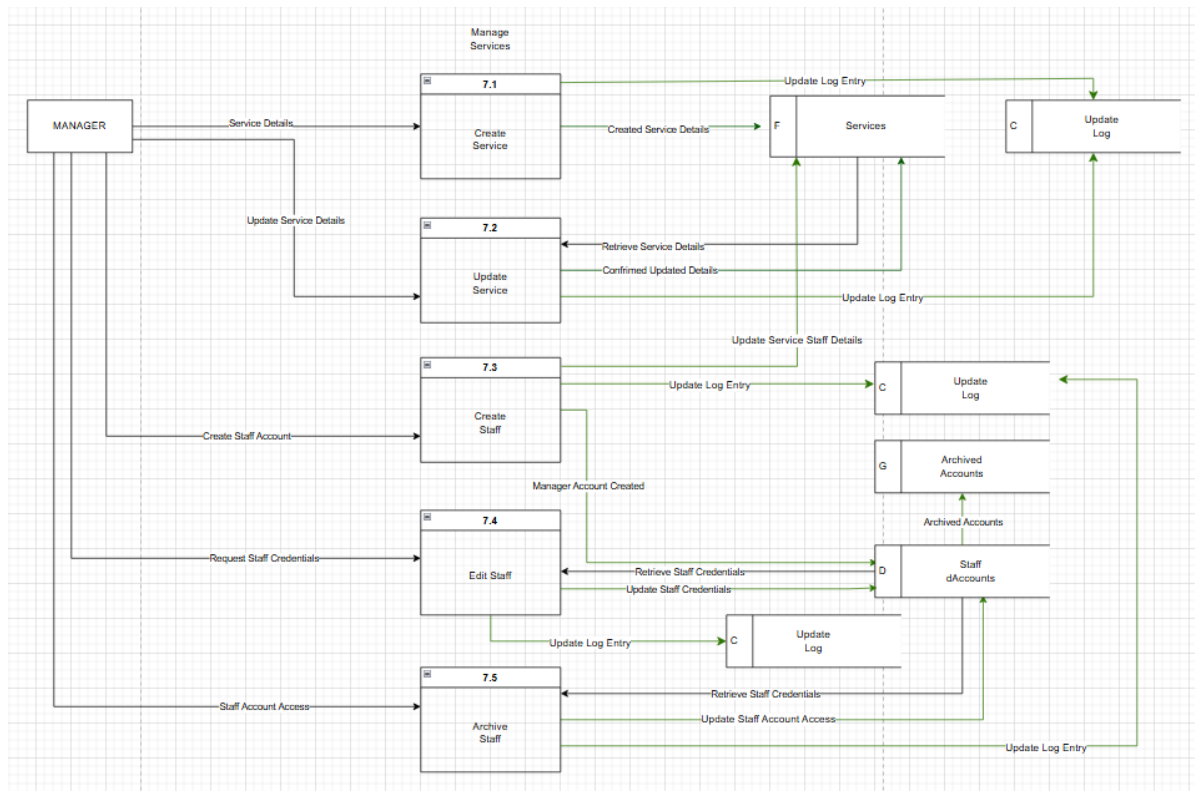


- Module 6: Manager Account Management





- Module 7: Service Management Window





5. Performance Requirements

In this, how a software system performs desired functions under specific conditions is explained. It also explains required time, required memory, maximum error rate, etc.

- **Response Time**

- API endpoints should respond within 2 seconds under normal load
- Authentication (login/logout) should complete within 3 seconds (accounts for bcrypt's intentional slowness at 12 rounds)
- Availability slot queries (getAvailableSlots) should return within 3 seconds

- **Throughput**

- The system should handle at least 20 concurrent users without crashing or throwing errors
- The appointment booking flow should complete within 5 seconds end-to-end

- **Error Rate**

- Critical API endpoints such as login, booking, status update, should have a success rate of at least 95% under normal conditions
- Email delivery should succeed on first attempt at least 90% of the time; failed emails must be queued and retried up to 3 times based on the EMAIL_MAX_RETRIES function

- **Availability**

- The backend server should remain up during a continuous 1-hour local dev session without requiring a manual restart
- The health check endpoint (GET /health) should always return 200 OK as long as the server process is running

- **Database**

- Database queries should complete within 2 seconds for typical reads (appointment lists, slot availability, audit logs)
- Soft-delete operations should complete within 1 second since they are single-row updates

- **Access Control**

- Every protected API route must resolve its RBAC check within 1 second
- Archived or inactive users must be denied access on every requestAdmin must have zero access to services and appointments endpoints by design (enforced in the permission matrix)

- **Authentication**

- Password reset tokens must expire within 1 hour and be invalidated immediately after a single successful use
- The system must reject login attempts with invalid credentials without revealing whether the email or password was wrong with the generic "Invalid credentials" message

- **Data Protection**



- The Supabase service role key must never be exposed to the client as it is server-side only and must be loaded exclusively from environment variables
- The audit_logs table must have no update or delete paths as it is insert-only by design
- Passwords must never be stored or logged in plain text and only bcrypt hashes are persisted
- Personal details in appointments.personal_details (JSONB snapshot) must not appear in server logs
- **Input Handling**
 - Password complexity must be enforced on registration, password change, and password reset with a minimum 8 characters, uppercase, lowercase, and a digit
 - Duplicate email registration must be rejected before any DB insert is attempted
 - Tracking number format must be validated against APT-YYYYMMDD-XXXXX before any lookup query is executed

6. Design Constraints

In this, constraints which simply means limitation or restriction are specified and explained for the design team. Examples may include use of a particular algorithm, hardware and software limitations, etc.

- **Functional Constraints**
 - The system must implement Role-Based Access Control (RBAC) for Admin, Manager, Staff, and Client roles.
 - The system must provide a web-based calendar interface with real-time appointment availability.
 - The system must ensure unique Appointment Tracking Numbers for every booking.
 - The system must validate all user inputs to prevent duplicates and invalid data.
 - The system must maintain audit logs for all actions related to users, appointments, and services.
- **Physical Constraints**
 - The system must run entirely in standard web browsers without requiring special hardware.
 - The system must display a responsive user interface that adapts to desktop, tablet, and mobile screen resolutions.
 - The system must use cloud-based database storage for all persistent data.
 - The system must rely on internet connectivity for real-time operations.
- **Performance Constraints**



- The system must respond to API requests within 2 seconds under normal load.
- The system must handle at least 20 concurrent users without errors or crashes.
- The system must complete end-to-end appointment booking within 5 seconds.
- The system must maintain at least 95% success rate for critical operations (login, booking, status updates).
- The system must enforce password complexity and validate credentials within 3 seconds.

- **Software Development Standards**

- The system must follow a client-server architecture separating frontend and backend logic.
- The system must use RESTful API design for communication between frontend and backend.
- The system must use consistent coding conventions, including naming, formatting, and modular design.
- The system must implement modular components for audit logging, notifications, and service management.
- The system must ensure maintainable, reusable, and extensible code for future upgrades.

7. Non-Functional Attributes

In this, non-functional attributes are explained that are required by the software system for better performance. An example may include Security, Portability, Reliability, Reusability, Application compatibility, Data integrity, Scalability capacity, etc.

Module 1: Security

- **NFA 1.1** The system shall hash all user passwords using bcrypt with a minimum cost factor of 12 before storing them in the database. Plaintext passwords shall never be stored or logged at any point
- **NFA 1.2** The system shall enforce Role-Based Access Control (RBAC) to ensure users can only access features and data permitted by their role (Admin, Manager, Staff, Client)
- **NFA 1.3** The system shall use JWT-based authentication with token expiration and refresh mechanisms to prevent unauthorized access.
- **NFA 1.4** The system shall use HTTPS encryption for all data transmission between client and server to protect sensitive information.

Module 2: Portability

- **NFA 2.1** The system shall support and be accessible to modern web browsers without loss of functionality.



- **NFA 2.2** The system shall be deployable on cloud platforms (e.g., Vercel and Supabase) without requiring major configuration changes.

Module 3: Reliability

- **NFA 3.1** The system shall maintain a minimum uptime of 99.5% (excluding pre-announced maintenance or downtime windows), ensuring that clients and staff can access appointment services at all operational hours.
- **NFA 3.2** The system shall handle errors and continue running without crashing when failures happen.
- **NFA 3.3** The system shall automatically retry failed email notifications to ensure message delivery.
- **NFA 3.4** The system shall implement database backups on a scheduled basis so that data can be restored to a recent state in the event of data loss or corruption.

Module 4: Reusability

- **NFA 4.1** The Audit Log System shall implement as a modular and independent component that can be reused in different parts of the system to record user actions without depending on one specific feature.
- **NFA 4.2** The Email Notification System shall implement as a modular component that can be reused for different actions like booking, status updates, and password reset.

Module 5: Application Compatibility

- **NFA 5.1** The system's backend API shall conform to REST architectural standards, ensuring that future third-party integrations can access its endpoints without requiring changes to the core system.
- **NFA 5.2** The system shall provide a responsive UI that supports various screen resolutions (for smooth responsiveness), including desktops, tablets, and mobile devices.

Module 6: Data Integrity

- **NFA 6.1** The system shall enforce referential integrity at the database level using foreign key constraints to keep data relationships correct.
- **NFA 6.2** The system shall validate all user input to make sure data is correct and avoid duplicate or invalid records.
- **NFA 6.3** The system shall implement audit logging to ensure all changes to data are traceable with timestamps and user identification.

Module 7: Scalability Capacity

- **NFA 7.1** The database shall be adaptable to at least 10 times the current data without major performance issues.
- **NFA 7.2** The system shall efficiently handle many users at the same time without slowing down or causing booking errors.



Module 8: Useability

- NFA 8.1 The system shall display clear, human-readable error messages for all failed operations (e.g., failed login, invalid form input, unavailable slot)

8. Preliminary Schedule and Budget

In this, initial version and budget of project plan are explained which include overall time duration required and overall cost required for development of project.

Major Task	Schedule	Budget
1. Planning		
1.1 Requirements Gathering	February 15 - 21, 2026	₱0
1.2 System & Database Architecture Design	February 15 - 21, 2026	₱0
1.3 UI/UX Mood Board & Prototyping	February 15 - 21, 2026	₱1,500
1.4 Project Management Tools	February 15 - 21, 2026	₱0
2. Documentation		
2.1 Technical Documentation (ERD, Data Dictionary)	February 15 - 21, 2026	₱0
2.2 Version Control Setup (GitHub free tier)	February 15 - 21, 2026	₱0
2.3 Final Report Printing & Binding	February 22 - May 23, 2026	₱1,500
3. Resource Preparation and Procurement		
3.1 Cloud Hosting Environment Setup	February 15 - 21, 2026	₱3,000
3.2 SMTP Email Service (notifications system)	February 15 - 21, 2026	₱1,500
3.3 Development Tools (VS Code,	February 15 - 21, 2026	



Node.js, libraries)		
4. Delivery and Setup		
4.1 Backend Setup & Configuration	February 15 - 28, 2026	₱6,000
4.2 Frontend Setup & UI Implementation	February 15 - 28, 2026	₱5,000
4.3 Full System Development	February 15 - May 2, 2026	₱18,000
4.4 System Integration (Frontend + Backend + Database)	April 19 - May 2, 2026	₱7,000
4.5 Database Deployment (Supabase)	April 19 - May 2, 2026	₱3,000
4.6 Hosting Deployment (Vercel / Hostinger)	April 19 - May 2, 2026	₱3,000
4.7 Security (SSL via Let's Encrypt, RBAC setup)	April 19 - May 16, 2026	₱2,000
4.8 Demo & Presentation Expenses	April 26 - May 9, 2026	₱1,500
5. Product Setup and Marketing		
5.1 User Guide Documentation	April 12 - 21, 2026	₱1,500
5.2 Client Coordination (meetings, communication)	April 19 - May 9, 2026	₱1,500
5.3 Project Kick-off	May 5 - 11, 2026	₱1,500
6. Monitoring and Assessment		
6.1 User Acceptance Testing	April 12 - May 9, 2026	₱1,000
6.2 Bug Testing & Fixing	April 12 - May 9, 2026	₱4,500
6.3 Performance Testing	April 29 - May 9, 2026	₱2,000



7. Project Closure		
7.1 Client Evaluation & Review	May 10 - 23, 2026	₱0
7.2 Certificate & Documentation	May 10 - 23, 2026	₱0
Grand Total:		₱65,000



9. Appendices

In this, additional information like references from where information is gathered, definitions of some specific terms, acronyms, abbreviations, etc. are given and explained.

9.1 Appendix A : Glossary of Terms

- API (Application Programming Interface) – A tool that lets two systems share data and work together.
- Audit Log – A record of actions done in the system.
- Booking Slot – An available date and time for an appointment.
- Client – The user or citizen who books an appointment.
- Dashboard – The main page where users can see information and options.
- Database – A place where system data is stored.
- JWT (JSON Web Token) – A secure token used to keep a user logged in.
- RBAC (Role-Based Access Control) – Access control based on the user's role.
- RESTful API – A way for systems to communicate using the internet.
- Supabase – The online service used for database and login features.
- Vercel – The online platform used to publish the website.

9.2 Appendix B: Acronyms and Abbreviations

- | | |
|---------|------------------------------------|
| ● API | Application Programming Interface |
| ● DB | Database |
| ● ERD | Entity Relationship Diagram |
| ● FR | Functional Requirement |
| ● HTTPS | Hypertext Transfer Protocol Secure |
| ● JSON | JavaScript Object Notation |
| ● JWT | JSON Web Token |
| ● NFA | Non-Functional Attribute |
| ● RLS | Row-Level Security |
| ● SMTP | Simple Mail Transfer Protocol |
| ● SSL | Secure Sockets Layer |
| ● UI | User Interface |
| ● UX | User Experience |



9.3 Appendix C: References

1. HubSpot. (n.d.-b). *Glossary of website terminology*. Retrieved from <https://blog.hubspot.com/website/glossary>
2. Postman. (n.d.). *API glossary: API & programming terminology*. Retrieved from <https://www.postman.com/api-glossary/>
3. Authgear. (n.d.). *Glossary*. Retrieved from <https://docs.authgear.com/reference/glossary>
4. Fieldset. (n.d.). *Glossary*. Retrieved from <https://fieldset.io/glossary>